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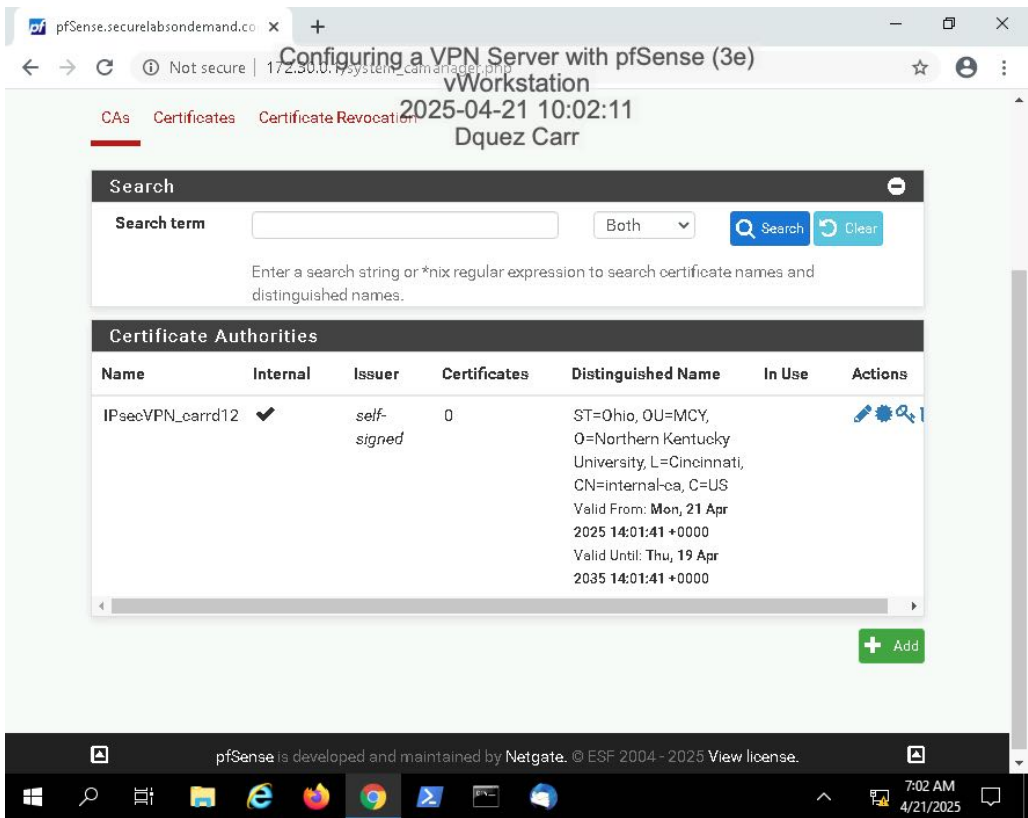
Time on Task:	Progress:
2 hours, 6 minutes	92%

Report Generated: Monday, April 21, 2025 at 10:56 AM

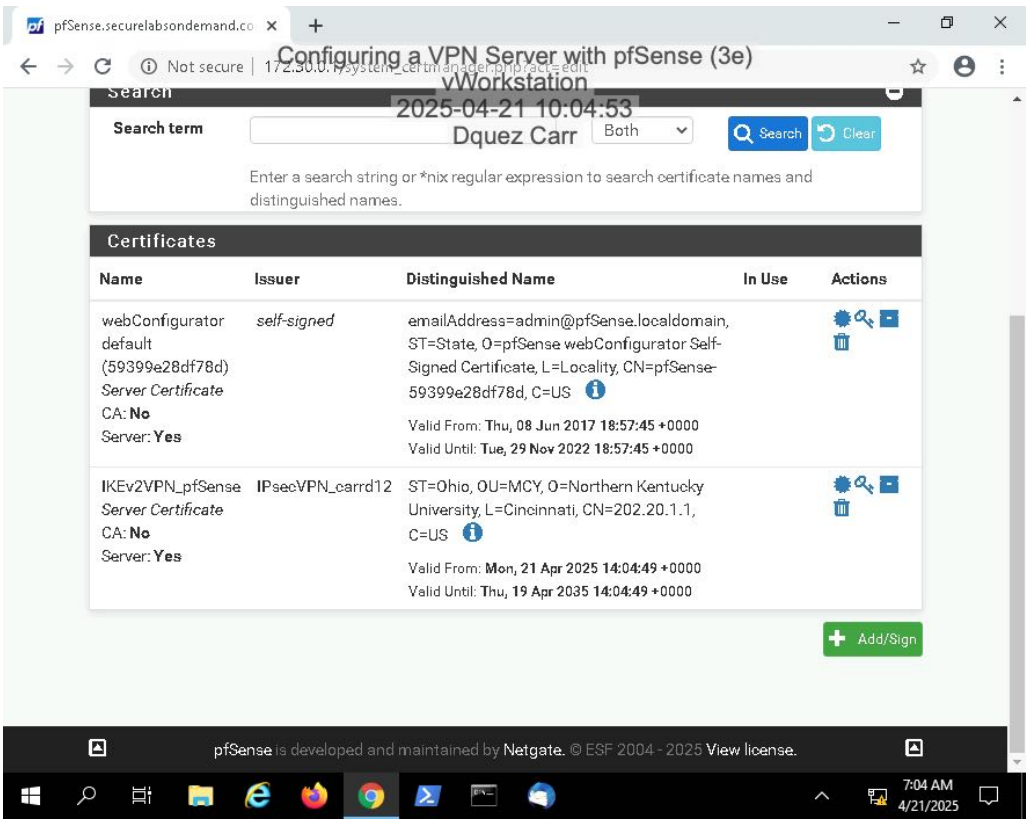
Section 1: Hands-On Demonstration

Part 1: Configure an IPsec VPN Server

18. Make a screen capture showing the updated Certificate Authorities table.



32. Make a screen capture showing the updated Certificates table.



Configuring a VPN Server with pfSense (3e)

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56. Make a screen capture showing the **updated IPsec Tunnels** table.

The screenshot shows the pfSense web interface for configuring IPsec tunnels. At the top, there is a warning message: "WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager." Below this, the breadcrumb navigation is "VPN / IPsec / Tunnels". The "Tunnels" tab is selected, and a success message states "The changes have been applied successfully." The main section is titled "IPsec Tunnels" and contains a table with the following data:

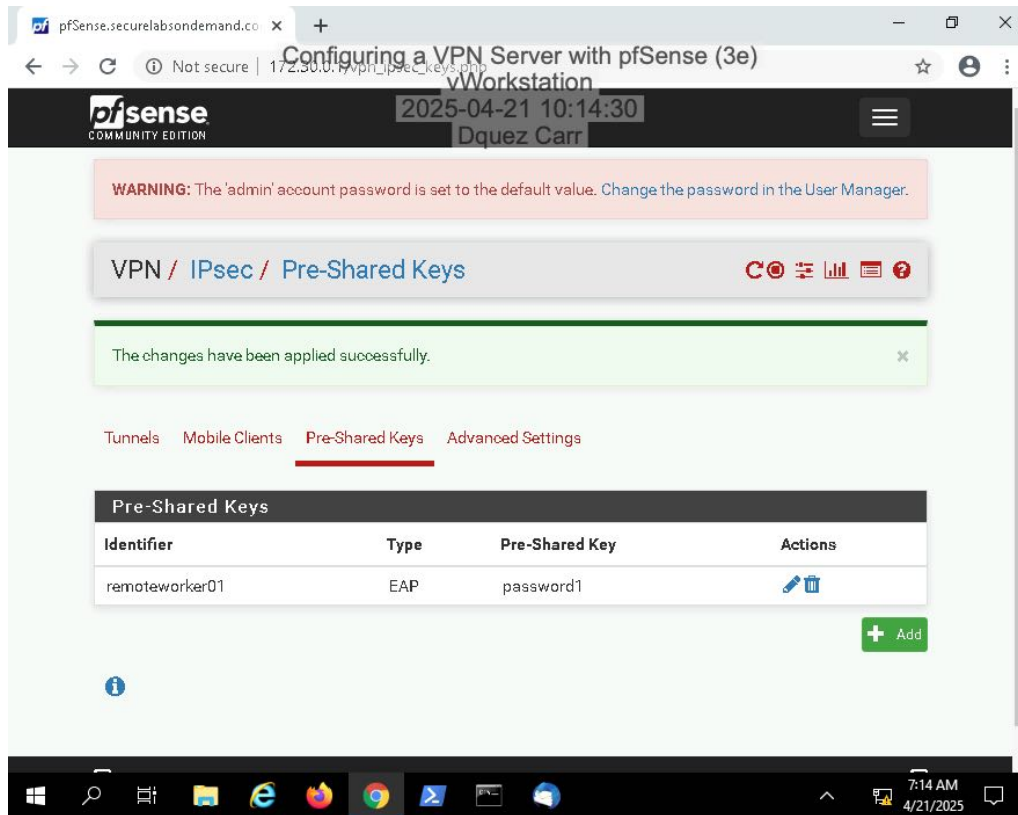
		Remote		P1	P1	P1 DH-	P1	
	IKE	Gateway	Mode	Protocol	Transforms	Group	Description	Actions
☐	Disable	V2	WAN Mobile Client	AES (128 bits)	SHA256	14 (2048 bit)	MobileIPsec	

Below the table is a button labeled "Show Phase 2 Entries (0)". At the bottom right, there are buttons for "Add P1" and "Delete P1s". The Windows taskbar at the bottom shows the time as 7:10 AM on 4/21/2025.

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74. Make a screen capture showing the **updated Pre-Shared Keys table**.

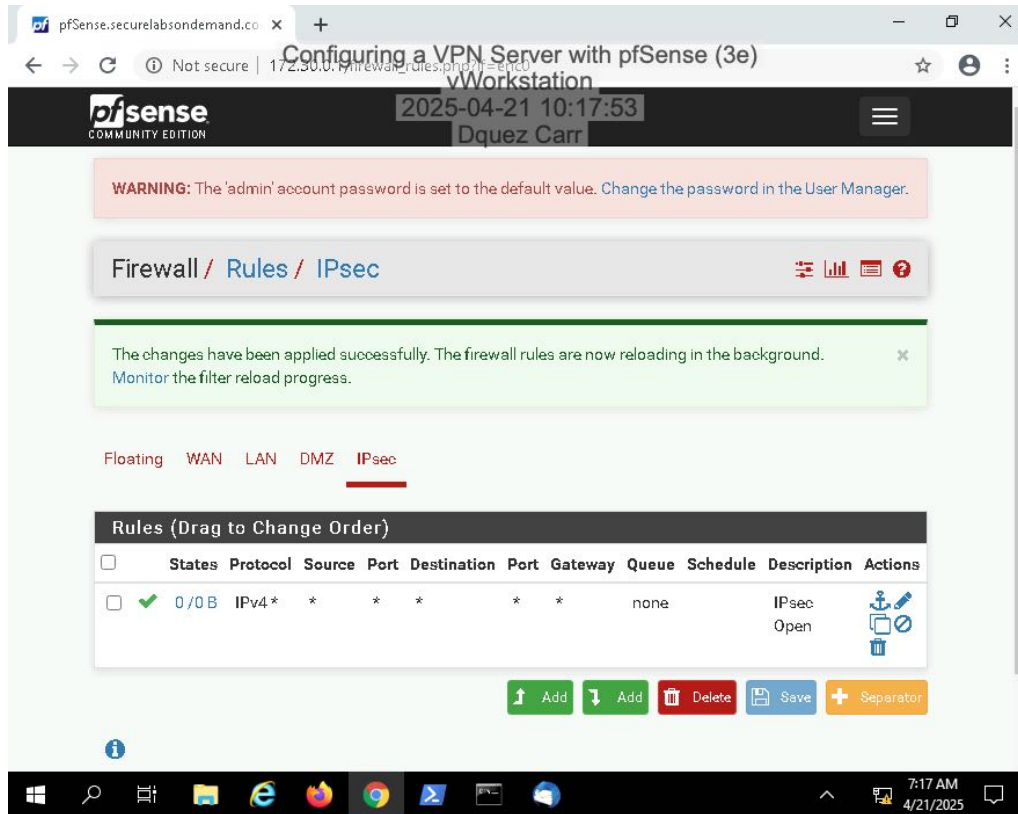


Part 2: Configure a Firewall Rule for VPN Traffic

Configuring a VPN Server with pfSense (3e)

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9. Make a screen capture showing the updated IPsec Rules table.



Section 2: Applied Learning

Part 1: Configure an OpenVPN VPN Server

13. Make a screen capture showing the CA configuration form.

The screenshot shows a web browser window with the title "Configuring a VPN Server with pfSense (3e)". The address bar shows "172.30.0.1/wizard.php?xml=openvpn_wizard.xml". The page content includes a text block explaining key size, followed by a form with the following fields:

- Lifetime**: 3650 (Lifetime in days. This is commonly set to 3650 (Approximately 10 years).)
- Country Code**: US (Two-letter ISO country code (e.g. US, AU, CA))
- State or Province**: Ohio (Full State or Province name, not abbreviated (e.g. Kentucky, Indiana, Ontario).)
- City**: Cincinnati (City or other Locality name (e.g. Louisville, Indianapolis, Toronto).)
- Organization**: Northern Kentucky University (Organization name, often the Company or Group name.)

At the bottom of the form is a blue button labeled "» Add new CA". The footer of the browser window states "pfSense is developed and maintained by Netgate. © ESF 2004 - 2025 View license." The Windows taskbar at the bottom shows the time as 7:20 AM on 4/21/2025.

Configuring a VPN Server with pfSense (3e)

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27. Make a screen capture showing the Tunnel Settings section.

The screenshot displays the 'Configuring a VPN Server with pfSense (3e)' wizard in a web browser. The 'Tunnel Settings' section is visible, showing the following configuration:

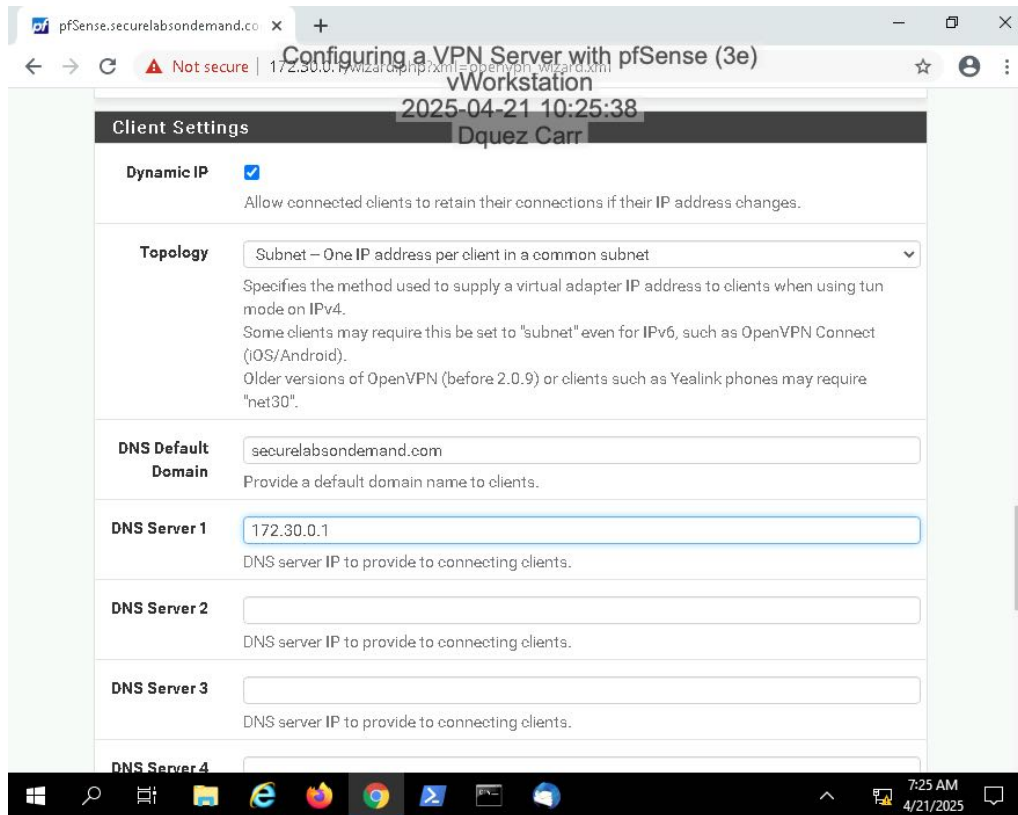
- Tunnel Network:** 172.31.1.0/24. Description: This is the virtual network used for private communications between this server and client hosts expressed using CIDR notation (eg. 10.0.8.0/24). The first network address will be assigned to the server virtual interface. The remaining network addresses will be assigned to connecting clients.
- Redirect Gateway:** ☐. Description: Force all client generated traffic through the tunnel.
- Local Network:** 172.30.0.0/24. Description: This is the network that will be accessible from the remote endpoint, expressed as a CIDR range. This may be left blank if not adding a route to the local network through this tunnel on the remote machine. This is generally set to the LAN network.
- Concurrent Connections:** 2. Description: Specify the maximum number of clients allowed to concurrently connect to this server.
- Compression:** Omit Preference (Use OpenVPN Default). Description: Compress tunnel packets using the LZO algorithm. Adaptive compression will dynamically disable compression for a period of time if OpenVPN detects that the data in the packets is not being compressed efficiently.
- Type-of-Service:** ☐. Description: Set the TOS IP header value of tunnel packets to match the encapsulated packet's TOS value.
- Inter-Client:** ☒

The Windows taskbar at the bottom shows the time as 7:24 AM on 4/21/2025.

Configuring a VPN Server with pfSense (3e)

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30. Make a screen capture showing the **Client Settings** section.

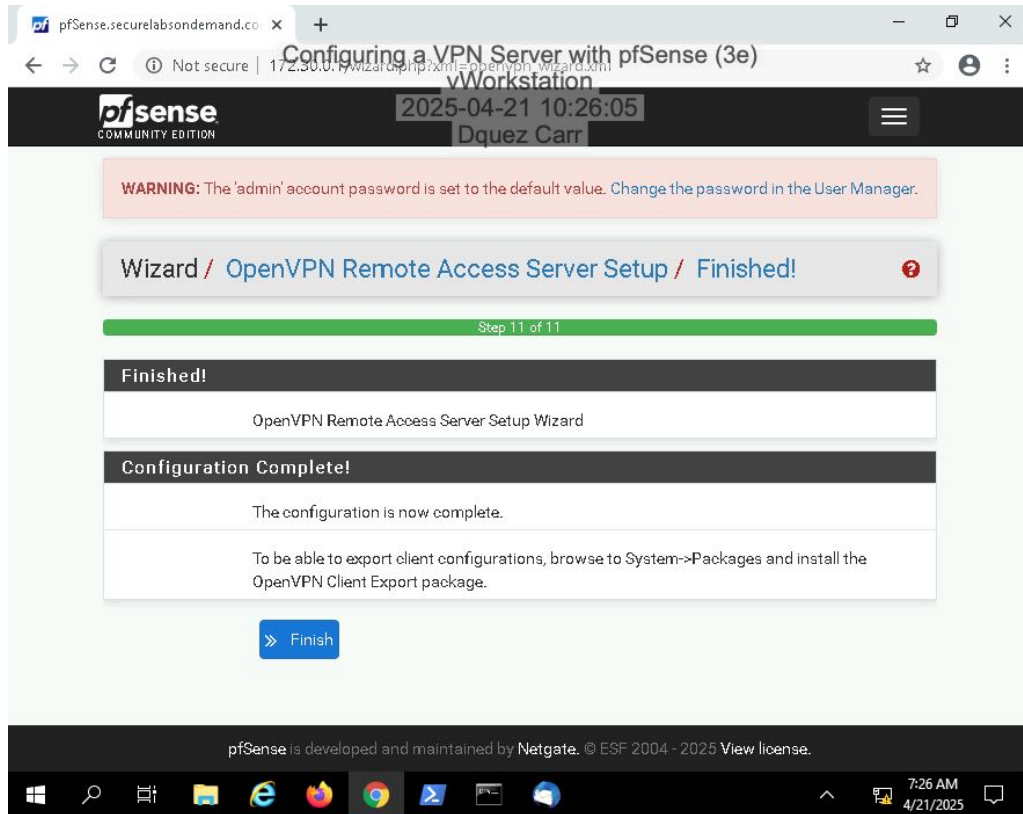


Part 2: Configure a Firewall Rule for VPN Traffic

Configuring a VPN Server with pfSense (3e)

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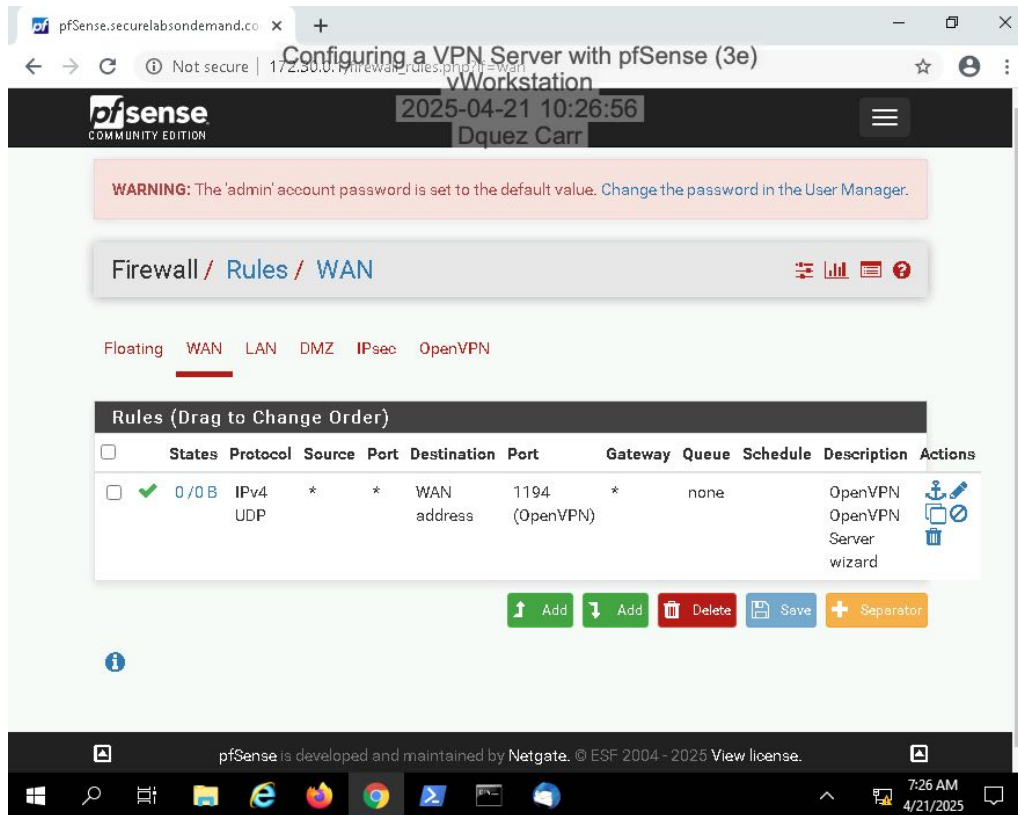
2. Make a screen capture showing the **completed OpenVPN configuration**.



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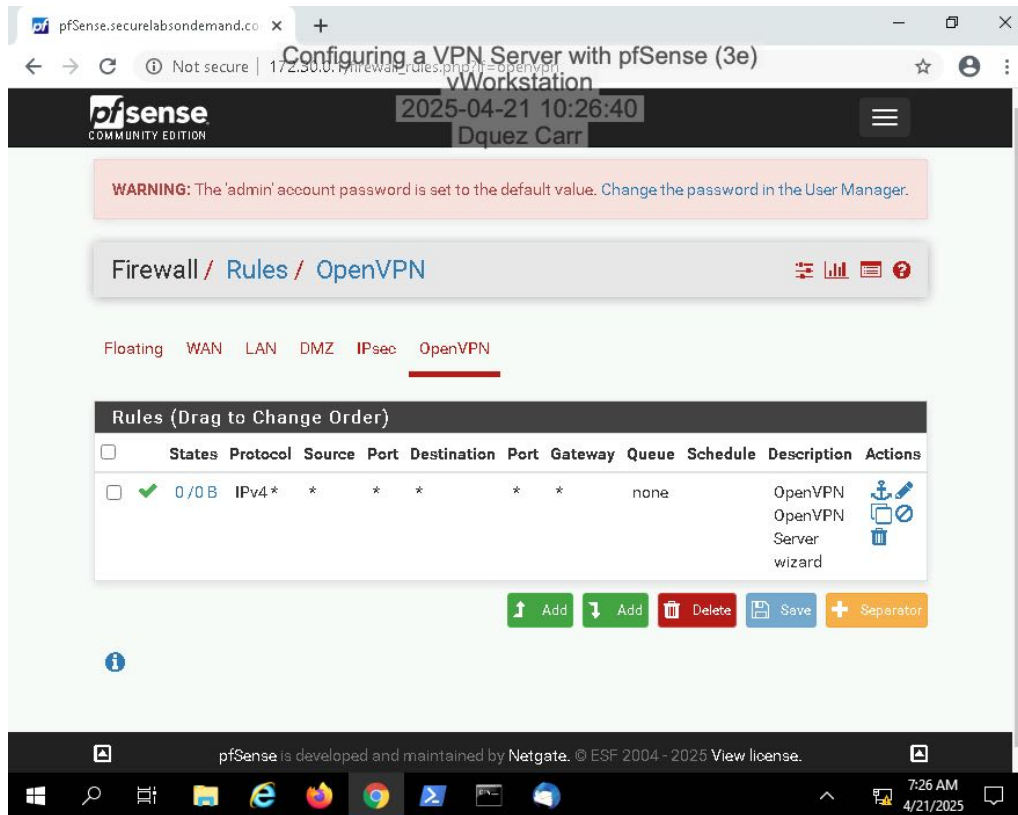
5. Make a screen capture showing the **OpenVPN** rule on the WAN Rules table.



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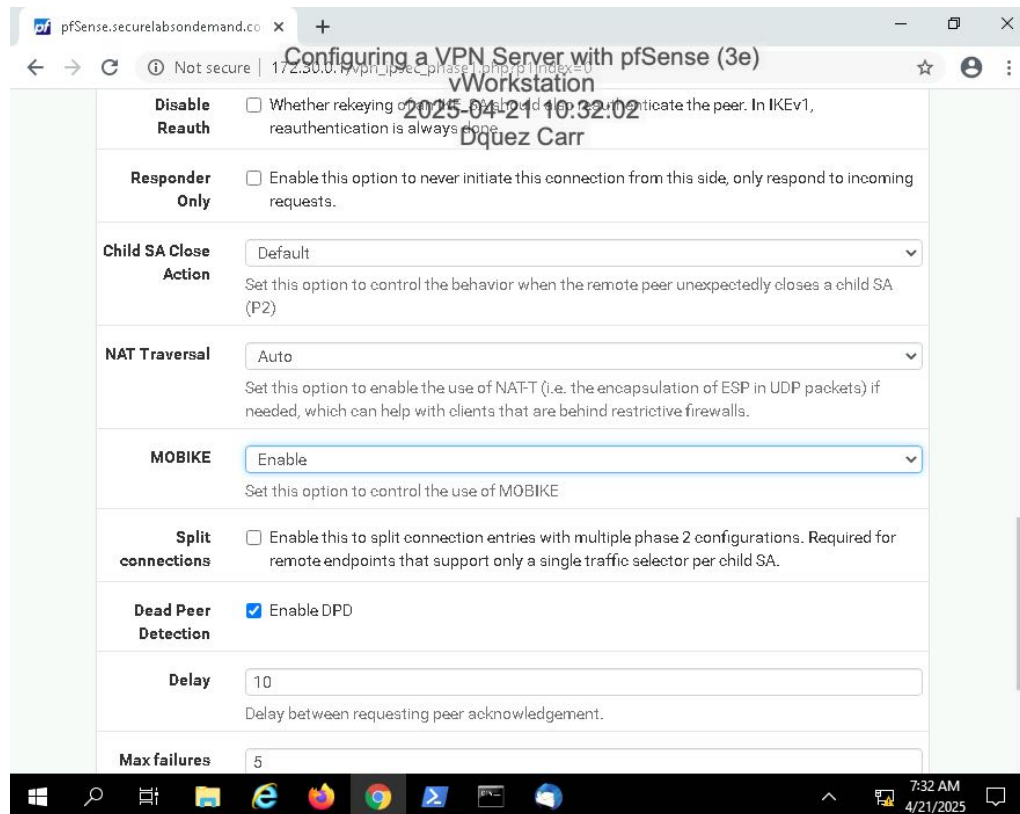
7. Make a screen capture showing the **OpenVPN** rule on the **OpenVPN Rules** table.



Section 3: Challenge and Analysis

Part 1: Enable IP Roaming for Remote VPN Clients

Make a screen capture showing the enabled **MOBIKE** option in the IPsec tunnel configuration.



Part 2: Create Explicit Firewall Rules for an IPsec VPN

Make a screen capture showing the **disabled** automatic IPsec rule creation option.

Incomplete

Make a screen capture showing your firewall rules that permit IPsec traffic.

